

Leo Chao

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EDUCATION

University of California, San Diego | La Jolla, CA

Sep 2026 - Present

Master of Science, Computer Science and Engineering

University of California, Irvine | Irvine, CA

Sep 2023 - Jun 2026

Bachelor of Science, Computer Science - Intelligent Systems (GPA: 3.96/4.00)

Honors: Outstanding Contribution to Research (Faculty Nominated), Dean's Honors List, Accelerated 3-Year Degree

SKILLS

Programming Languages: Python, C++

LLM & Agents: Google ADK, Vertex AI, PyTorch, HuggingFace Transformers, LangGraph, LLM-as-a-Judge pipelines

Interpretability & Evaluation: TransformerLens, SAELens, Sparse Autoencoders (SAE), Activation Steering

Tools: GCP, Docker, Git, Linux, FastAPI

RESEARCH EXPERIENCE

Julian McAuley Lab - UC San Diego | La Jolla, CA, *Research Collaborator*

May 2026 - Present

- Collaborating with Xin Xu on mechanistic interpretability of music LLMs using sparse autoencoders and TransformerLens; targeting ICLR 2027

ARCALA Research Lab - UC Irvine | Irvine, CA, *Research Assistant*

Jan 2025 - Jun 2026

- Co-developed Chat-Bloom, an LLM evaluation taxonomy derived from Bloom's taxonomy, to classify and measure behavioral shifts in human-LLM interaction logs; organized and analyzed a dataset of 8,076 prompt-response pairs
- Trained BERT/DistilBERT ensemble classifiers for automated interaction taxonomy labeling, achieving ~90% accuracy validated against human-coder baseline (75.9% inter-rater reliability)
- **Chao, L.**, et al. "Chat-Bloom: A Taxonomy for Classifying Cognitive Offloading in Student-AI Interactions in CS Education." Accepted, SIGCSE Virtual 2026

PROFESSIONAL EXPERIENCE

LSEV USA Corp | Irvine, CA, *AI Intern*

Jun - Aug 2026

- Designed a quality filter that decides which LLM-extracted product claims stay published, using cross-family judging and source re-grounding, reaching 96% agreement with human labels across a 300-product set labeled by two annotators
- Reduced published claim errors by 65% across 2,444 product pages by splitting validation into separate evidence-existence and sentiment-correctness checks
- Built a 3-layer query classifier resolving 80% of product searches deterministically under 40ms at zero API cost, cutting LLM spend by 82% versus judging every query

Acer | Taipei, Taiwan, *AI Systems Intern*

Jun - Aug 2025

- Architected a multi-agent SDLC automation system using Google ADK and Vertex AI, reducing manual iteration time by 35%; deployed to GCP with a production path into internal Google Agentspace
- Refactored LangGraph prototype into production-grade ADK workflow with persistent memory management, integrating 6 enterprise GCP services including Cloud Storage and Firestore
- Designed state management architecture for long-horizon agent tasks, ensuring context preservation across 10+ dynamic subtask agents executing code generation, API stub creation, and test drafting

TECHNICAL PROJECTS

LLM Activation Steering | Mechanistic Interpretability (*TransformerLens, SAELens, PyTorch*)

Jan - Mar 2026

- Built an inference-time intervention pipeline to inject SAE feature vectors into Gemma's residual stream, evaluating sentiment and truthfulness steering across 1,372 IMDb and 817 TruthfulQA samples
- Designed an LLM-as-a-Judge evaluation framework using GPT-5 with multi-dimensional metrics: relative sentiment shift, atomic truthfulness verification, and normalized coherence scoring
- Found myth-conviction features unexpectedly improved truthfulness scores, suggesting frequency-based selection captured entangled refusal proxies rather than pure semantic targets